# Final Internship Project :

# PROJECT REPORT

**Name :** Lekshmy Sajeevan Sindhu

**Course :** Web Development Course (June batch)

# Module 1:
# Web Technology Fundamentals

Related Topics:

● Introduction to Web Technologies ● Client-Server Architecture ● Types of Websites

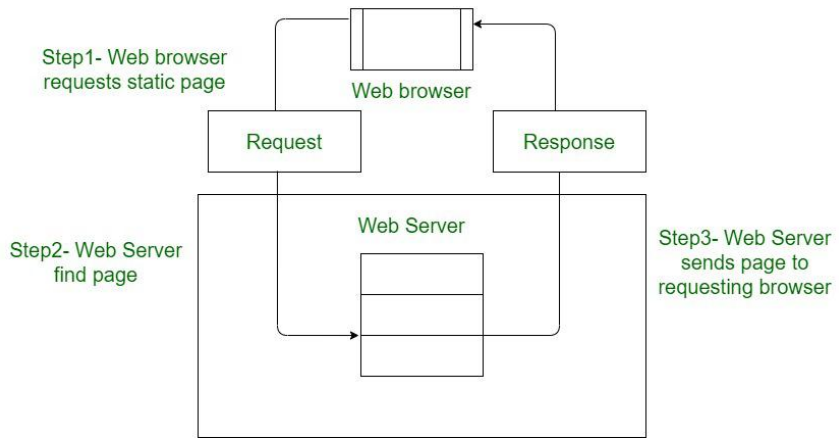
Web workflow

A website design workflow is a structured, step-by-step process that guides a project from the initial idea to launch and ongoing maintenance

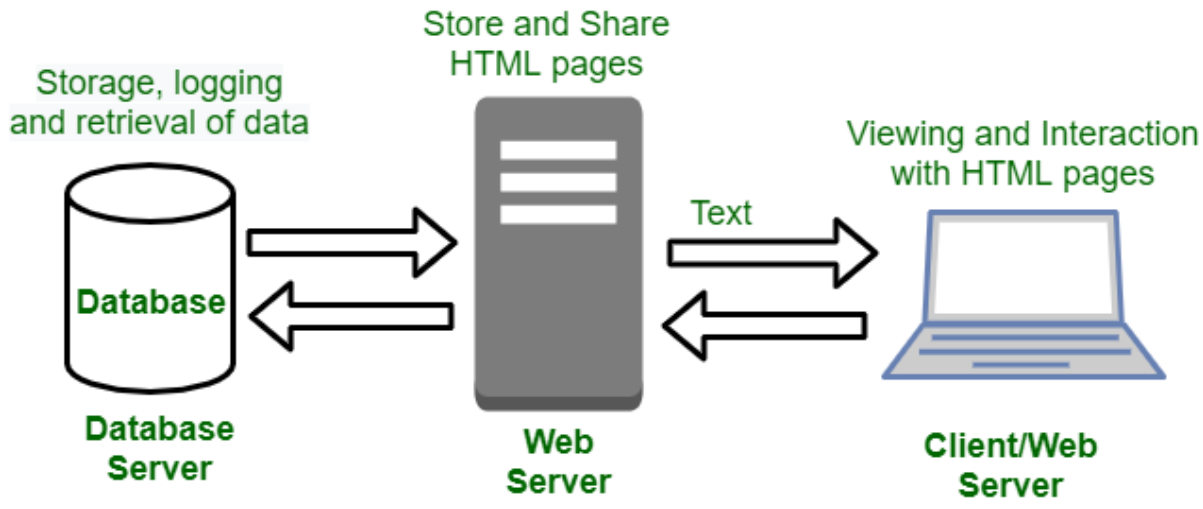
When a user interacts with a website, data flows through a **three-tier architecture** consisting of the Frontend, Backend, and Database.

- **Frontend (The Client):** The user triggers an action (e.g., clicking "View Profile"). The browser packages this request into an **HTTP Request** (containing headers, a method like GET or POST, and sometimes a body) and sends it over the internet to the server.
- **Backend (The Server):** The server receives the request, validates the user's session, executes business logic, and determines what data is needed. It speaks to the database using database-specific drivers or languages (like **SQL**).
- **Database (The Storage):** The database processes the query, fetches or updates the requested rows/documents, and sends the raw data back to the backend.
- **The Return Trip:** The backend formats this data (usually into **JSON** or HTML) and sends it back to the frontend as an **HTTP Response** (with a status code like 200 OK). The browser renders the update for the user.

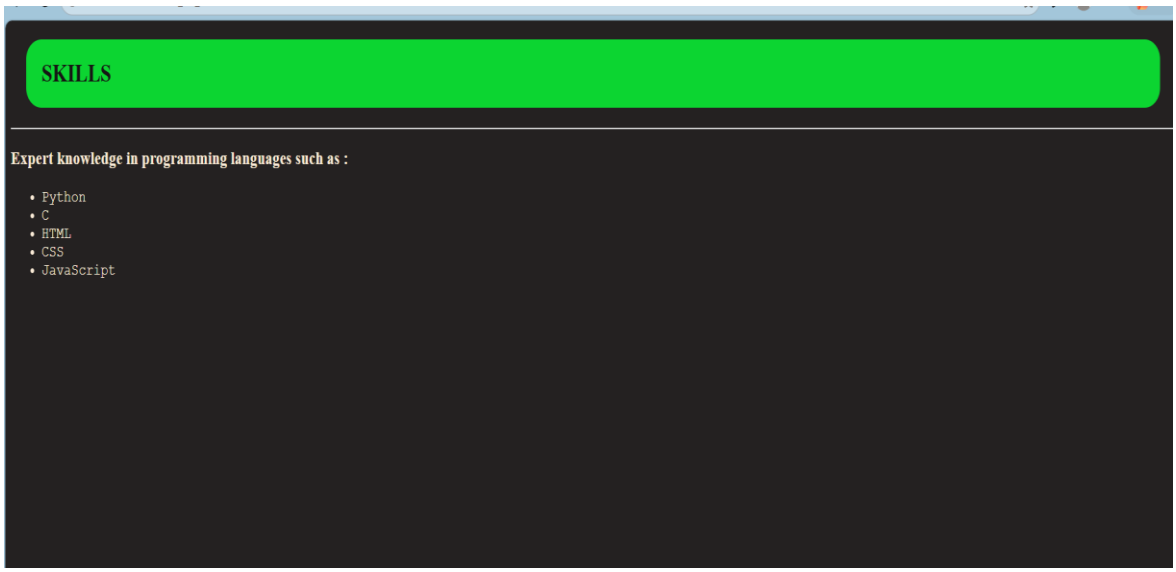
Web pages are returned by the server which are prebuilt source code files built using simple languages such as HTML, CSS, or JavaScript. There is no processing of content on the server (according to the user) in Static Websites. Web pages are returned by the server with no change therefore, static Websites are fast. There is no interaction with databases. Also, they are less costly as the host does not need to support server-side processing with different languages.



In Dynamic Websites, Web pages are returned by the server which are processed during runtime means they are not prebuilt web pages but they are built during runtime according to the user's demand with the help of server-side scripting languages such as PHP, Node.js, ASP.NET and many more supported by the server. So, they are slower than static websites but updates and interaction with databases are possible.




Basic webpage :



# Module 2:
# Frontend – HTML

Related Topics:

- HTML Elements, Forms, Lists, Tables Tasks:
- Adding Images, Links, and Special Characters

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**LinkedIn :**

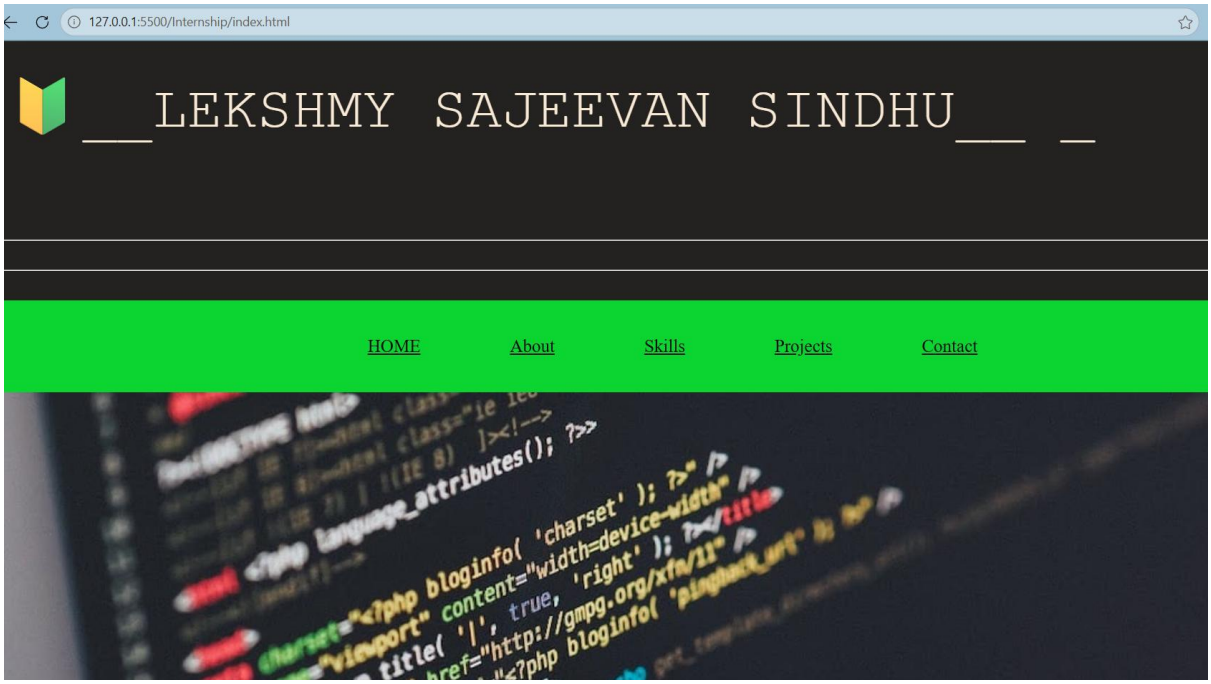
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**GitHub :**

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# Module 3:
# Styling – CSS

Related Topics:

- CSS Selectors, Colors, Fonts, Box Model
- Responsive Layouts

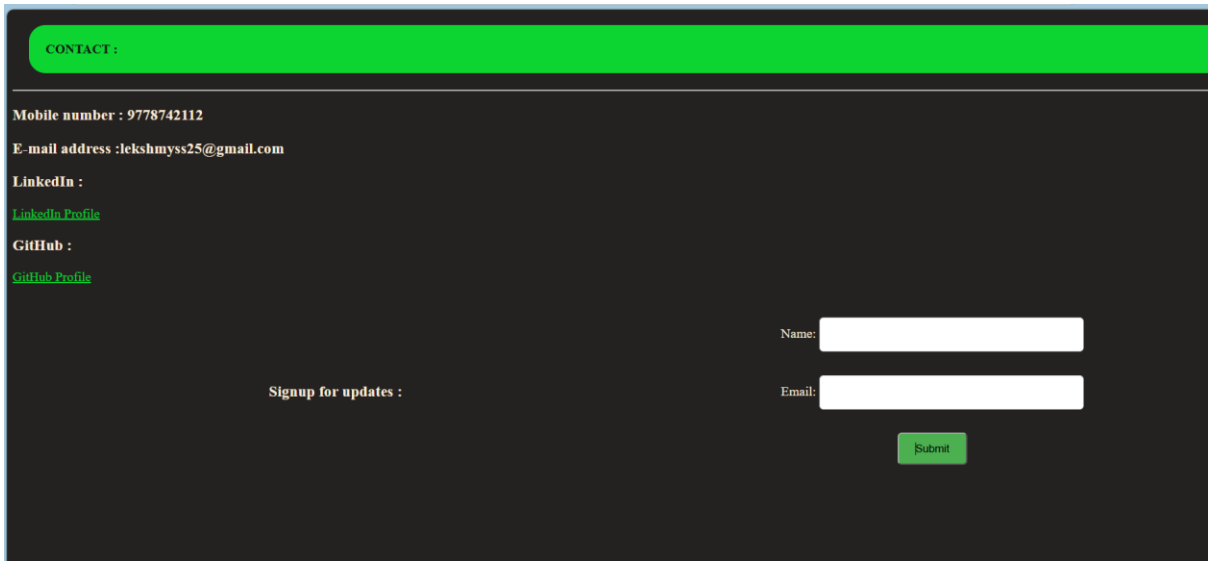





# Module 4 :
# Dynamic Scripting – JavaScript

Related Topics:

- Variables, Loops, Events, Functions, Cookies
- Page Redirection

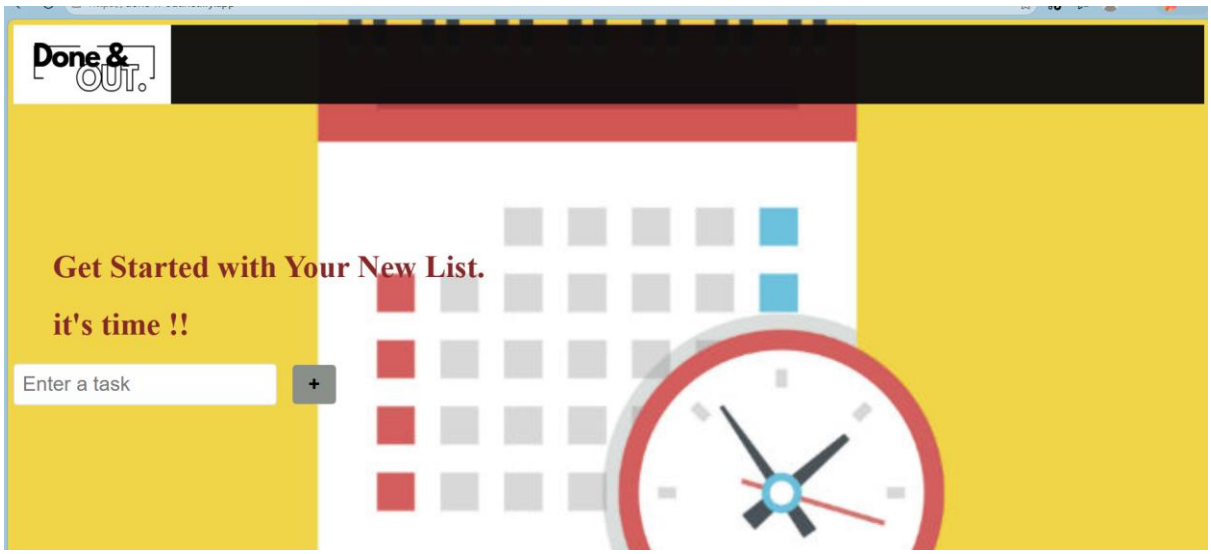


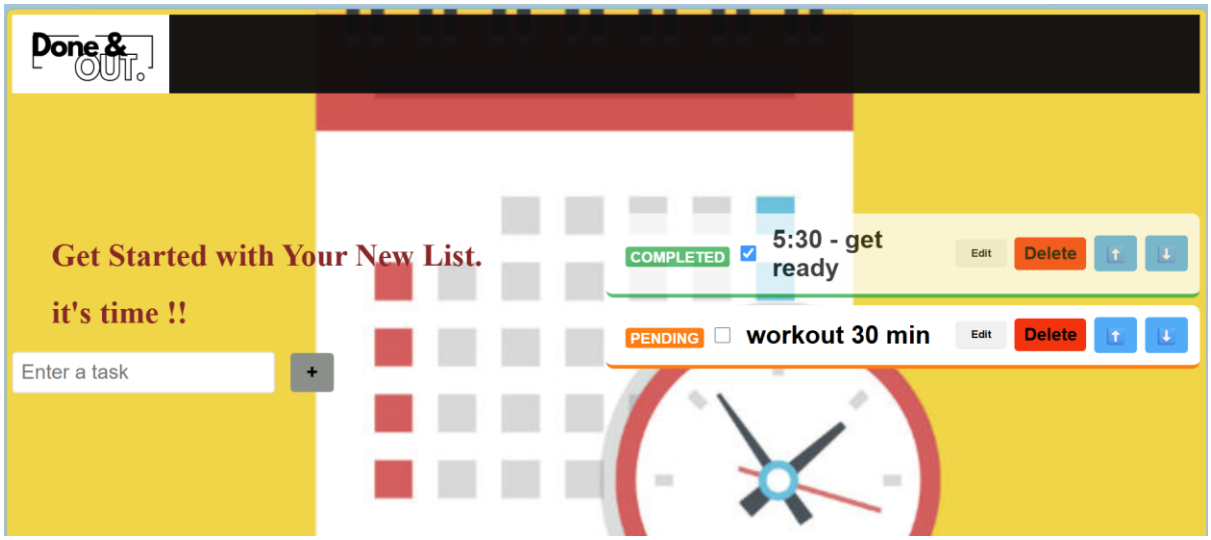



# Module 5 :
# Advanced JavaScript & DOM

Related Topics:

- DOM Manipulation
- Advanced Form Validation

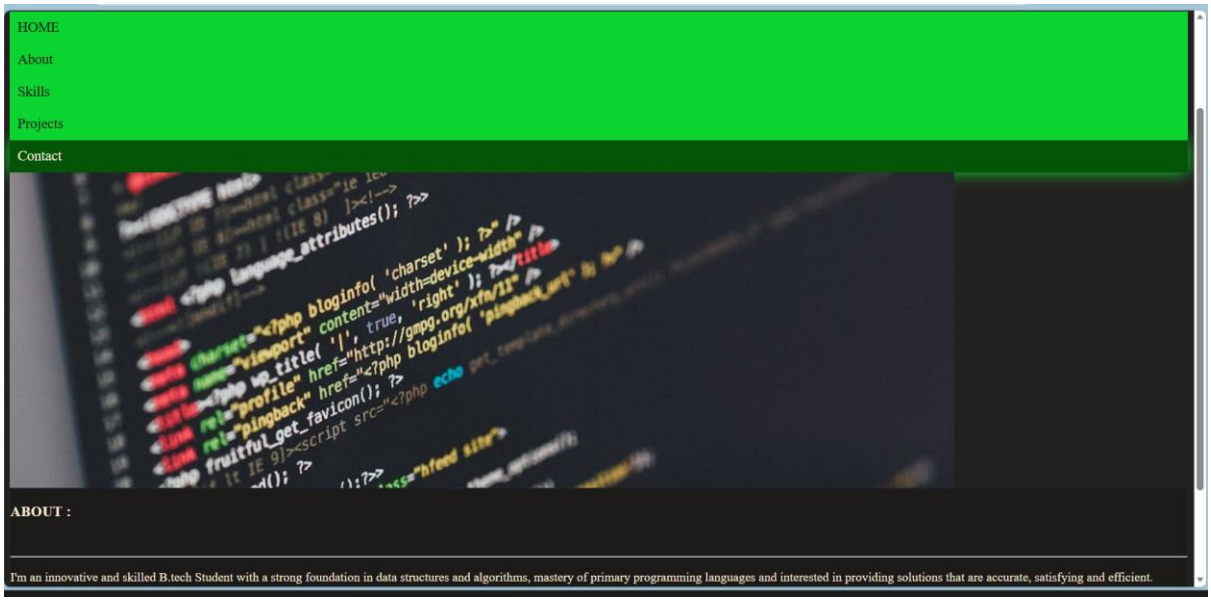


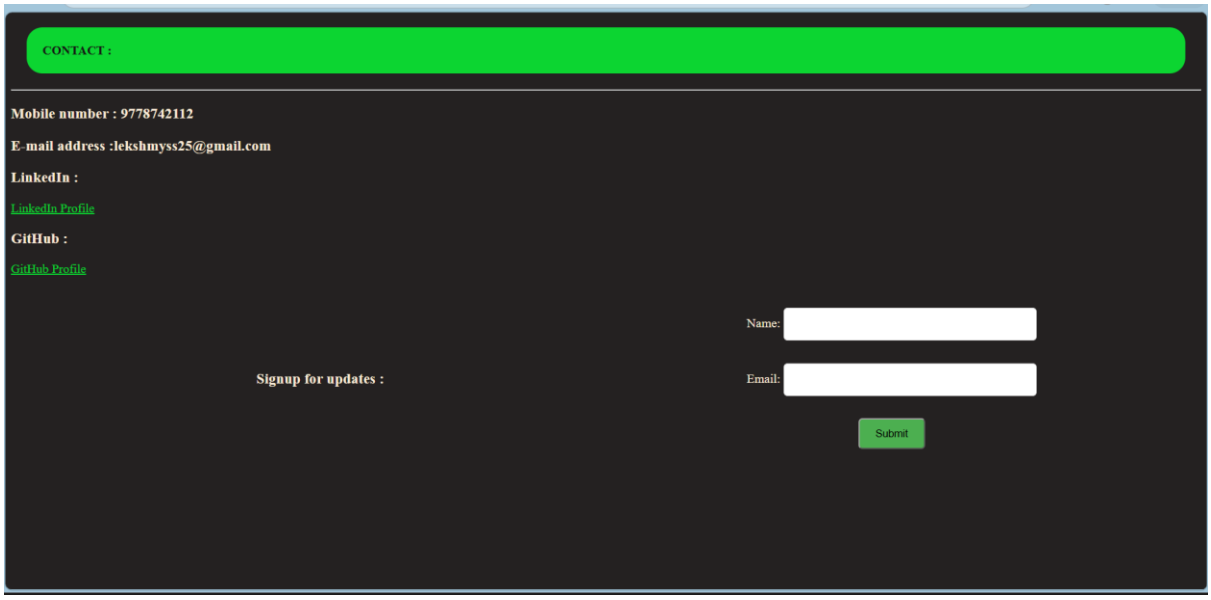



# Module 6 :
# Bootstrap Framework

Related Topics:

- Grid System, Typography, Buttons, Modals, Navbars

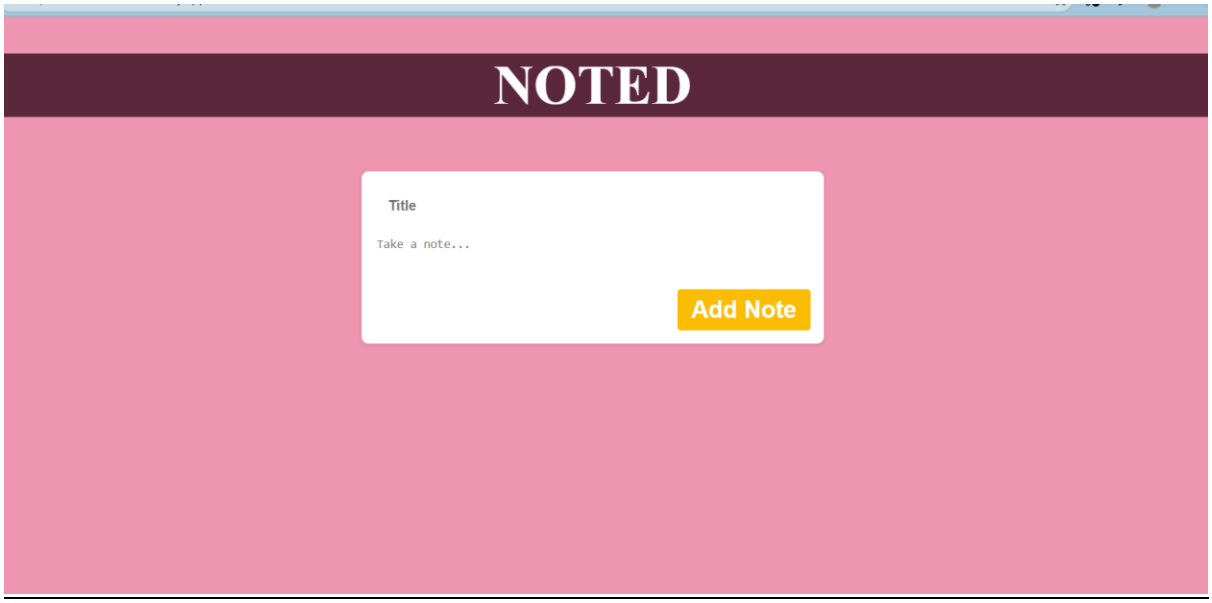


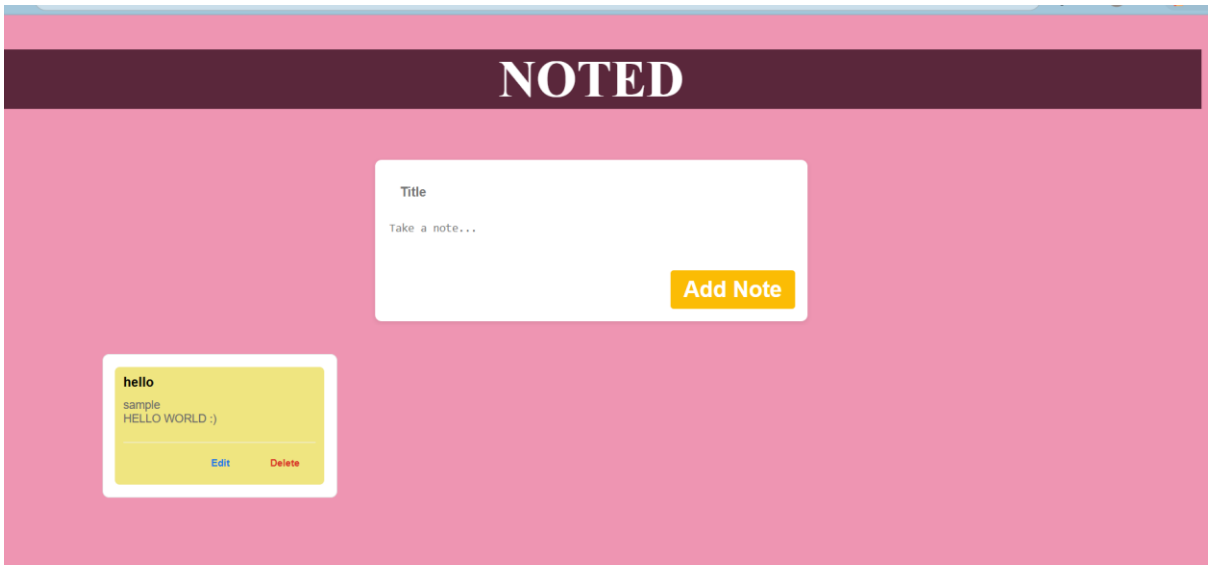



# Module 7 :
# React Basics

Related Topics:

- Components, Props, Forms, Project Structure

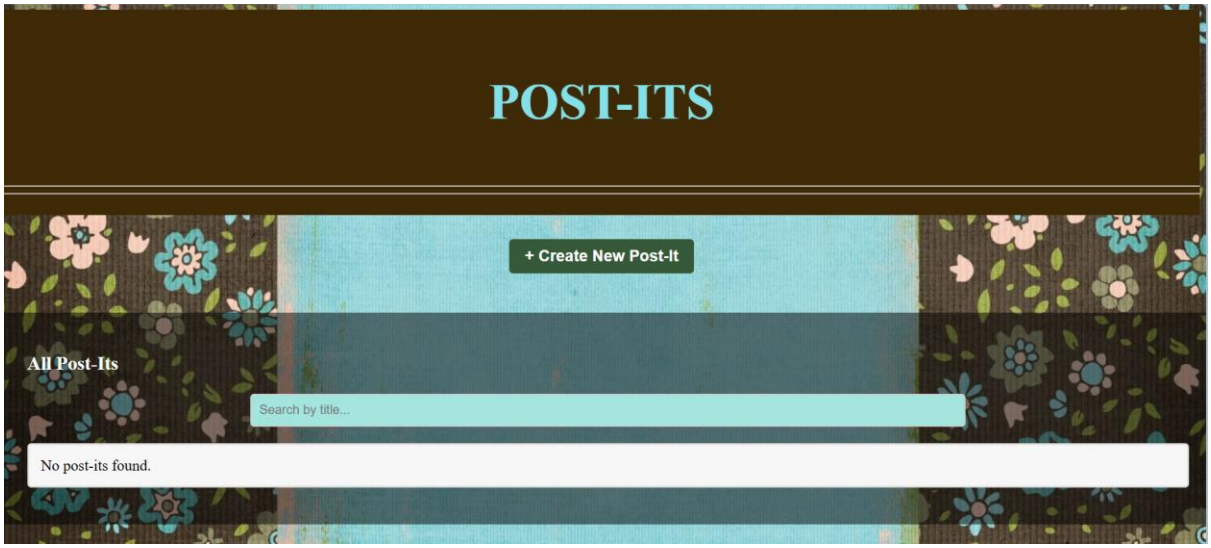


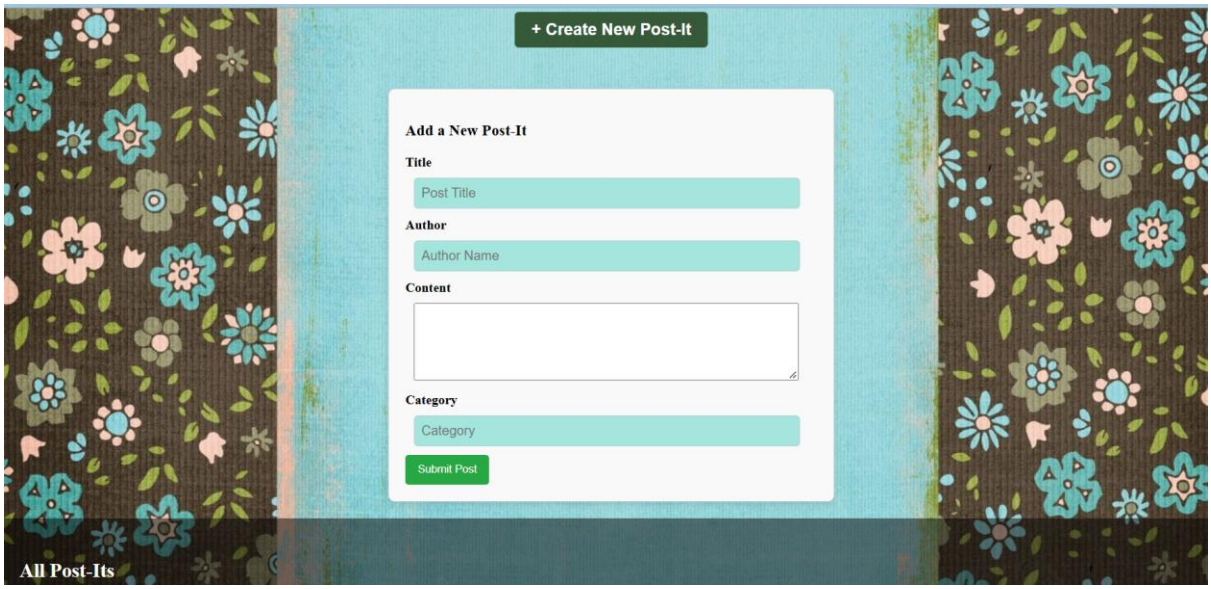



# Module 8 :
# React Advanced

Related Topics:

- API Integration, State Management, Testing



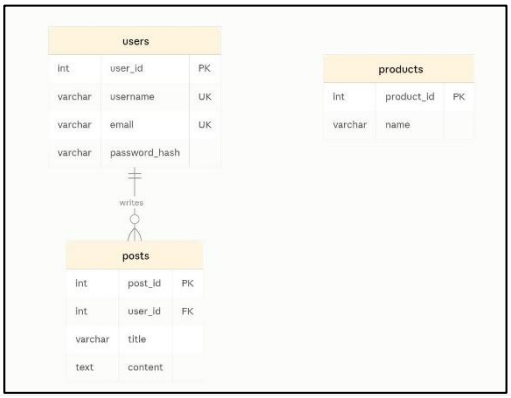



# Module 9 :
# Databases

Related Topics:

- MySQL & MongoDB
- CRUD Operations and Joins

Database Schema-




```sql
CREATE TABLE users (
    user_id INT PRIMARY KEY,
    username VARCHAR(50) NOT NULL UNIQUE,
    email VARCHAR(100) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL
);

CREATE TABLE products (
    product_id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL
);


CREATE TABLE posts (
    post_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    title VARCHAR(150) NOT NULL,
    content TEXT NOT NULL,
    FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
);

insert into users values(1,"ABC","ab@12gmail.com","12xx");
insert into users values(2,"XYZ","xy@89gmail.com","89aa");

insert into products values(101, "watch");
insert into products values(102, "notes");

insert into posts values (1,1,"HELLO","xxx");
insert into posts values (2,1,"sample","hello_world");
select * from users;
select * from products;
select * from posts;
```

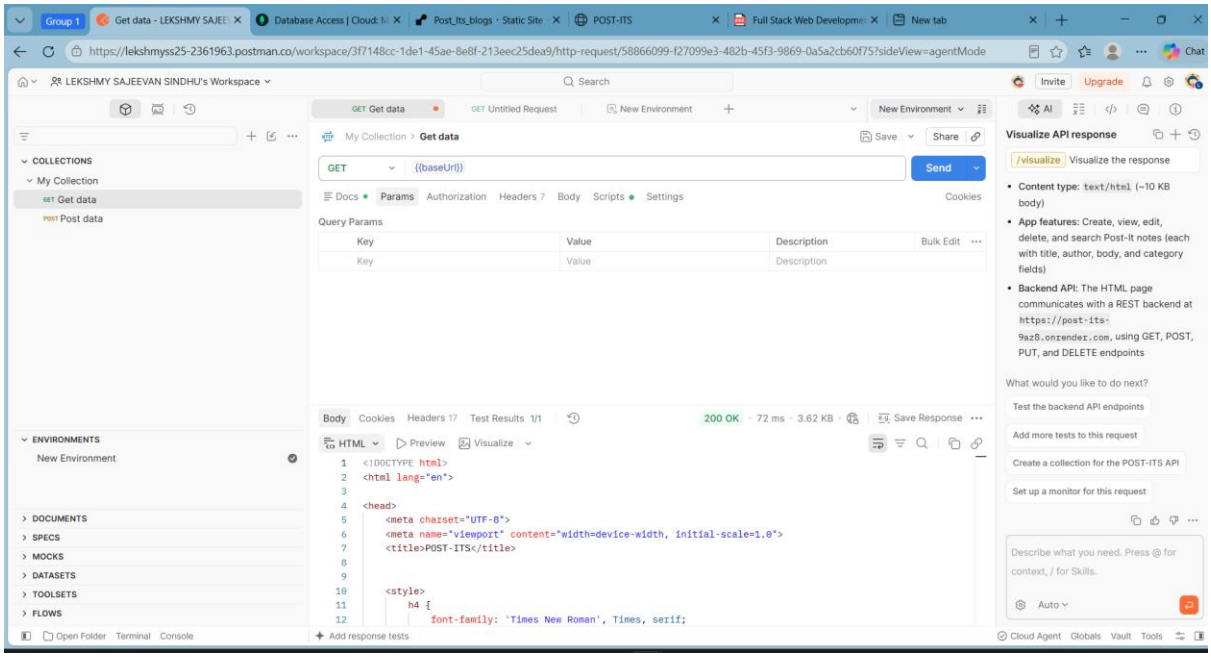
Output:

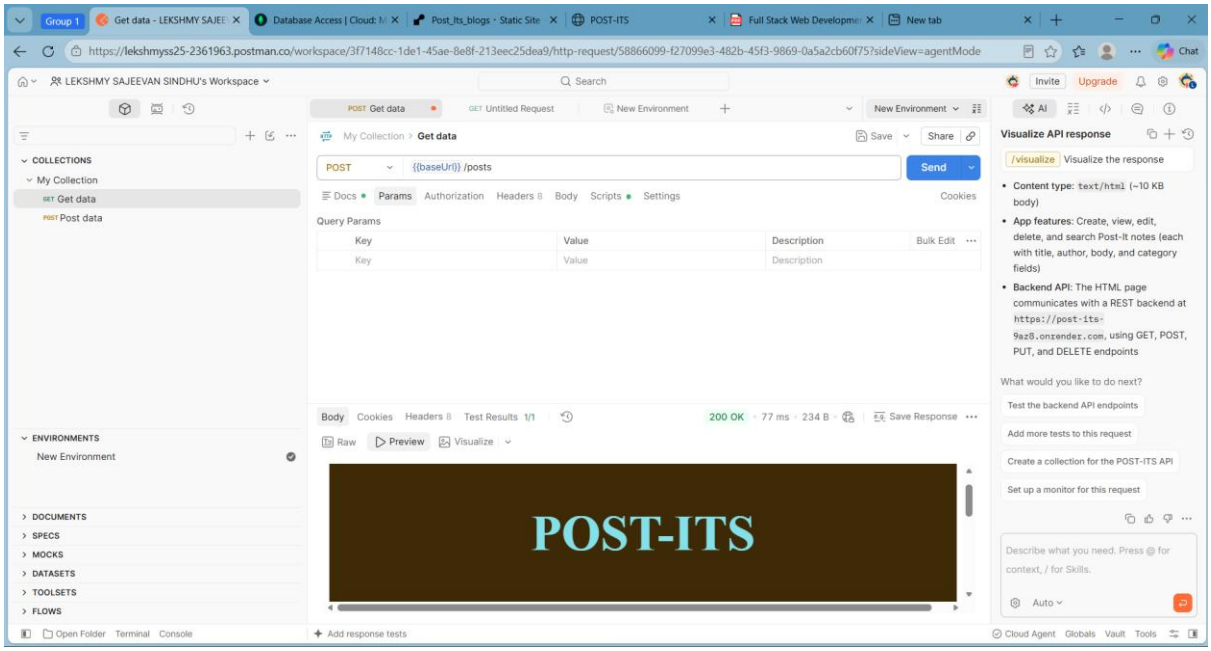
```
+---------+----------+----------------+---------------+
| user_id | username | email          | password_hash |
+---------+----------+----------------+---------------+
|       1 | ABC      | ab@12gmail.com | 12xx          |
|       2 | XYZ      | xy@89gmail.com | 89aa          |
+---------+----------+----------------+---------------+
+------------+-------+
| product_id | name  |
+------------+-------+
|        101 | watch |
|        102 | notes |
+------------+-------+
+---------+---------+--------+-------------+
| post_id | user_id | title  | content     |
+---------+---------+--------+-------------+
|       1 |       1 | HELLO  | xxx         |
|       2 |       1 | sample | hello_world |
+---------+---------+--------+-------------+
```

# Module 10 :
# Backend – Node.js & Express.js

Related Topics:

- Express Routing, Middleware, APIs



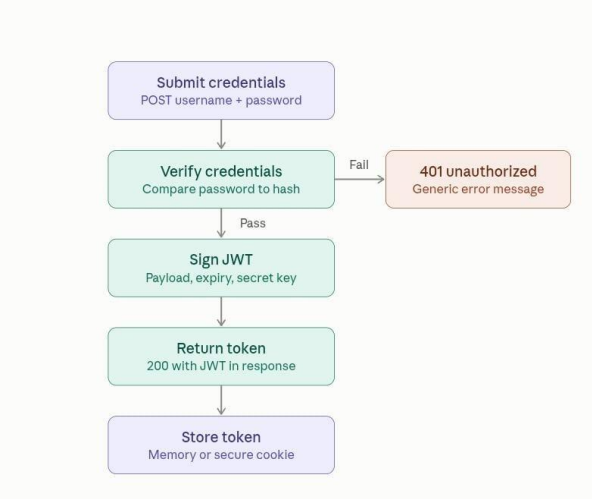



# Module 11 :
# Authentication & Deployment

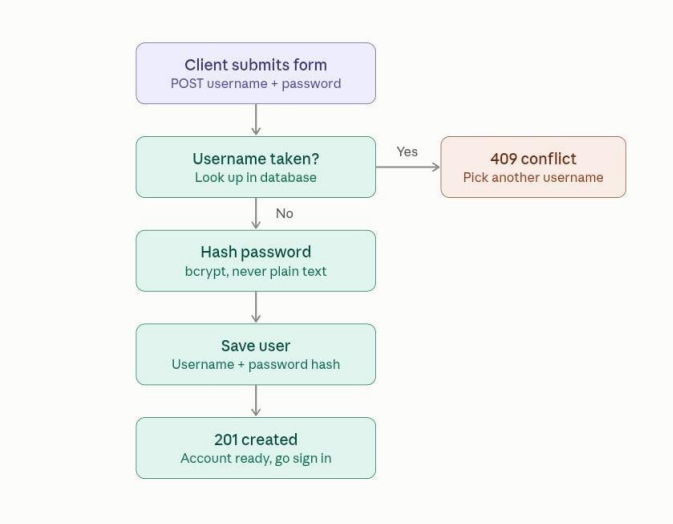
Related Topics:

- JWT, bcrypt, Middleware
- Deployment on Render/Netlify

Deployed link : https://lekshmy-ss-portfolio.onrender.com




SIGN IN




SIGN UP

# Module 12 :
# Final Full Stack Project

Related Topics:

- Live Website Design & Integration

Deployed link : https://lekshmy-ss-portfolio.onrender.com

GITHUB : blog_render/Internship at main · lekshmyss25-ayw/blog_render